

1|ONELASER

VertiGoTM

**Performance Rotary Laser for Drinkware
(38W RF)**



Specializing in high-margin cylinders

Zero hassle, instant setup to double your output.

Double Your Throughput

Featuring the world's first vertical motion system, it shatters the bottlenecks of traditional horizontal engraving. Driven by an 800mm/s top speed and 3G maximum acceleration, it doubles your Units Per Hour (UPH) output, empowering you to dominate high-margin markets like thermoses and tumblers.

Zero Hassle, Instant Setup

Built with a fully integrated rotary axis and true auto-focus technology. Simply load your cup and start with a single button. Ready to run in just 15 minutes out of the box, it completely eliminates tedious mechanical alignments and manual focusing errors, putting an end to costly scrap rates.

Production-Grade Consistency

Powered by a 38W RF CO2 metal tube and a precision gantry system. Regardless of the coating color, the laser delivers 100% consistent, perfectly perpendicular output from the rim to the base. It guarantees 1,000 DPI photo-quality, distortion-free engraving that traditional Galvo systems simply cannot match.

The World's First Vertical Laser for Tumblers

VertiGo is the first laser machine purpose-built for engraving cylindrical objects like tumblers, bottles, dog bowls, and even massive ice buckets, with stunning precision and zero hassle. Say goodbye to complicated rotary setups and hello to easy engraving.



STANLEY TUMBLER



BEER STEIN



WINE BOTTLE



DOG BOWL



VASE



ICE BUCKET



COFFEE MUG



ROLLING PINS



PROMOTIONAL ITEMS



GLASS WATER CUP



THERMOS



YETI MUGS

10 Key Innovations

01 Revolutionizing Cylinder Engraving

VertiGo's vertical design transforms cylinder processing, giving your studio a competitive edge with cutting-edge production methods.

02 Precision Beyond Gravity

The vertical motion system ensures flawless acceleration and accuracy, maintaining smooth, consistent engraving during high-volume production.

03 Zero Hassle, Instant Setup

The integrated rotary axis offers a "plug and play" setup—no adjustments required, allowing operators to start creating immediately.

04 Unmatched Speed & Performance

With 3G acceleration and 800mm/s engraving speed, VertiGo doubles output, increasing production and revenue.

05 Unrivalled Detail, Flawless Precision

1,000DPI resolution captures every detail, delivering sharp, precise engravings that elevate product quality.

06 Next-Level Power

Powered by a 38W RF CO2 metal laser tube, VertiGo ensures consistent, high-quality engraving with a longer lifespan and lower operating costs.

07 Effortless Control at Your Fingertips

The full-touch control system simplifies operation, with real-time updates and minimal training required.

08 Instant Visualization, Seamless Execution

The WYSIWYG touchscreen interface perfectly matches the vertical system, reducing learning time and enhancing workflow efficiency.

09 Irregular surface compatibility

Shape-Shifting versatility: VertiGo adapts to irregular shapes like cups and baseball bats, expanding business opportunities and giving you a competitive edge.

10 Automatic focusing system

Automatic focus, perfect results every time. The automatic focusing system guarantees precision, reducing errors and waste with each batch.

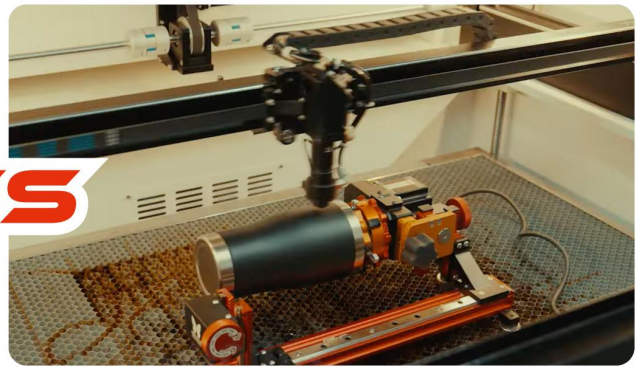


Innovation-Driven Performance

Ditch the "Toys" and Embrace "Production-Grade" Throughput



Vertical Engraving



Horizontal Engraving



EASY, UPRIGHT ENGRAVING
(Quick & accurate)



NO GRAVITY ISSUES
(Secure for all sizes)



COMPLEX SETUP & ALIGNMENT
(Costly mistakes)



FIGHTING GRAVITY
(Heavy/bulky items shift)



FULLY PORTABLE
(Trade shows & events)



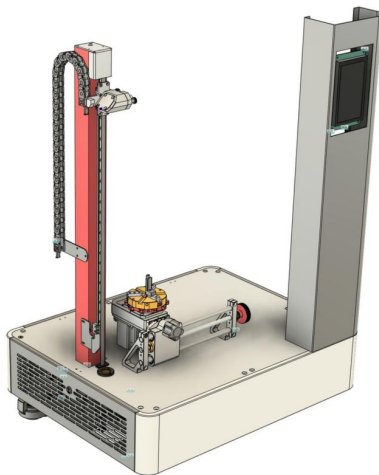
FAST, LIVE ENGRAVING
(High earning potential)



NOT PORTABLE
(Stuck in shop)



SLOW WORKFLOW
(Kills creativity)



The Magic of Vertical Gravity

Shattering the Limits of Traditional Horizontal Engraving Our exclusive vertical motion system defies gravity, breaking through the bottlenecks of traditional horizontal flat engraving.

It achieves a blazing top engraving speed of 800mm/s and a maximum acceleration of 3G.

It's not just about raw speed—it's about delivering an exceptionally high "Units Per Hour" (UPH) output.

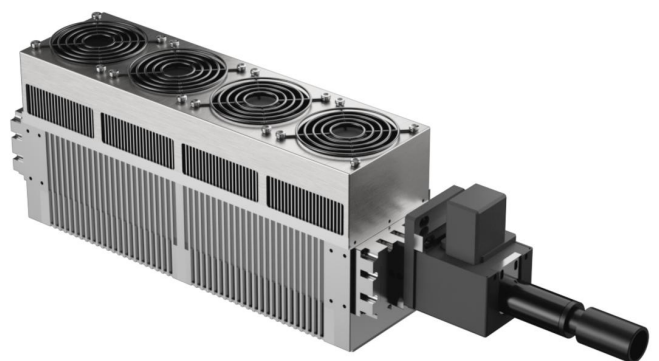
Why a 38W RF CO2 Gantry System?

Compared to UV lasers, whose microscopic spot size makes large-area filling agonizingly slow, or Diode lasers that cannot process clear glass and fail on light-colored coatings, VertiGo's 38W RF CO2 metal tube is the ultimate production engine for thermoses and powder-coated tumblers.

The gantry architecture ensures that the laser maintains 100% consistent, perfectly perpendicular output and a flawless circular spot size from the rim of the cup all the way to the base. It completely rejects the edge distortion, elongation, and stitching seams that often plague traditional Galvo systems.

1,000 DPI Photo-Quality Resolution

Delivering razor-sharp edge quality that meets the highest commercial standards, ensuring your finished products command a higher premium.



Outcome Guarantee

Zero Learning Curve, Zero Scrap

Out of the Box to Profitable in 15 Minutes

Arriving 99% pre-installed, VertiGo is operational in just 15 minutes. It features built-in air cooling and exhaust systems, completely removing the need for clunky external chillers. Plug it in, and you are minutes away from producing your first billable order.

"Zero Hassle" Rotary Setup

The integrated rotary chuck is built directly into the machine frame with zero wobble, delivering true "plug-and-play" performance. Say goodbye to the nightmare of manually squaring the rotary or dialing in steps-per-rotation (SPR) for every single tumbler.

True Auto-Focus & One-Touch Operation

Simply load your cup, press a button, and start engraving. The machine automatically and precisely pinpoints the perfect focal distance, totally eradicating manual measurement errors and putting an end to costly material waste.



Smart Resume (Power Failure Protection)

A sudden power loss no longer means tossing an expensive blank into the trash. When power is restored, the smart resume feature flawlessly picks up the exact remaining progress, minimizing downtime and safeguarding your production.

The Money Maker

Capture the Most Lucrative 30% of Your Revenue



Dominate High-Margin Markets

Data reveals that tumbler engraving accounts for nearly 30% of the income for many engraving shops. VertiGo empowers you to specialize in and dominate this highly lucrative segment.

Shape-Shifting Versatility

Perfectly compatible with a wide array of objects ranging from 89mm to 293mm in diameter. Whether it's the trending Stanley tumbler with a handle or a solid baseball bat, VertiGo tackles them effortlessly, instantly expanding your product SKUs and business opportunities.

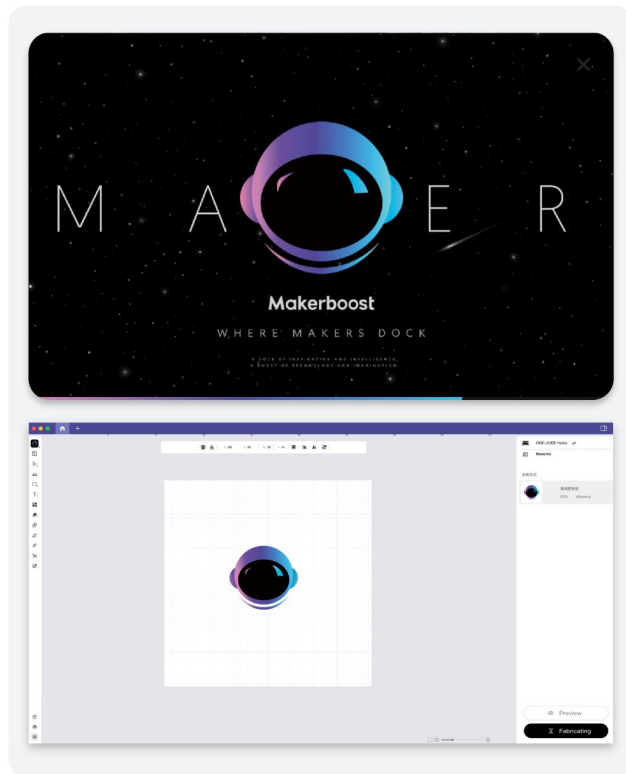
The Portable "Money Printer" (Instant Customization)

Featuring a compact footprint (630x550x710mm), ultra-quiet 70dB noise reduction design, and a built-in air cooling system, VertiGo allows you to bring the machine directly to trade shows, weekend markets, and retail counters. Deliver "Instant Customization, Instant Profit" right on the spot.

Makerboost

Built by OneLaser. Optimized for VertiGo.

Makerboost goes beyond mere compatibility — it's fully integrated into the VertiGo platform, with functions designed exclusively for high-efficiency engraving, batch coding, and intelligent alignment. It provides operators with robust tools through a streamlined interface and automates tasks that previously demanded additional steps.

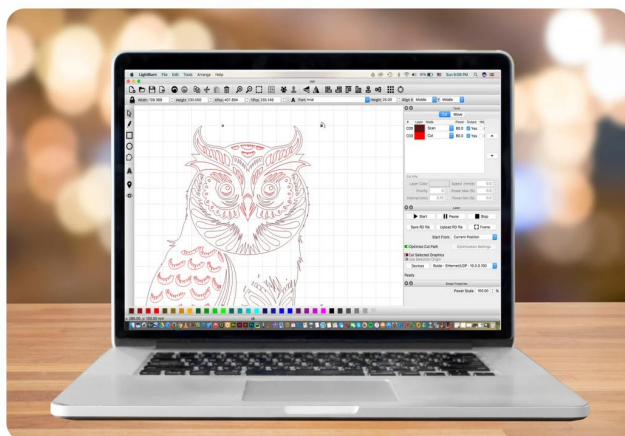


Key Advantages of Makerboost on VertiGo:

- Auto Device Detection: Automatically identifies the VertiGo model and laser power, and applies the correct parameters instantly.
- Mark Point Recognition: Automatically detects registration marks to ensure accurate alignment for engraving applications.
- Smart Variable Data Support: Effortlessly generate barcodes, QR codes, serial numbers, and dynamic text from CSVs or templates for efficient batch engraving.
- Advanced Output Optimization: Fine-tune images, add lead-in/out lines, and apply kerf offset to achieve precise fits and professional engraving results.

Makerboost + VertiGo = Quick setup, fewer mistakes, seamless workflow — ready to go instantly.

LightBurn



Compatible with LightBurn and supported formats: AI, PDF, SC, DXF, HPGL, PLT, RD, SCPRO2, SVG, LBRN, BMP, JPG, JPEG, PNG, GIF, TIF, TIFF, and TGA

VertiGo Specs

Machine Size	
Large Workpiece Capacity	11.81"(300mm) in height and 11.54"(293mm) in diameter
Machine Size	24.8"x 21.65"x 27.95" (630x550x710mm)
Machine Weight	141.10 lb (64kg)
Laser Source	
RF Metal Tube	38W RF CO2 metal laser tube
RF Metal Cooling Method	Air Cooling
Performance	
Max Acceleration	3 G
Max Speed	800 mm/s
Positioning Accuracy	<=0.000394"(0.01 mm)
Max Scanning Precision	1,000 DPI
Machining Size (Auto focus)	Diameter 89~230mm / 3.5"~9.05"
Machining Size (Manual focus)	Diameter 16~230mm / 0.62"~9.05"
Configuration	
Red Light Indication	✓
Auto Focus	✓
Smoke Extraction	✓
Optics	
Beam Combiner	3X
Protective Window	2.5"
Lens Diameter	20mm / 0.79"
Mirror Diameter	25mm / 0.98"
Power Requirements	
Voltage	110 V / 220 V
Electricity Requirement	100-240V AC, 50 or 60Hz, Single phase Machine requires either 115V or 220V voltage(customized from factory); Support both 50Hz and 60Hz, single phase
Power Consumption	706.2W, 115V 660.5W, 220V
Support	
Knowledge Base	✓
Warranty	✓ (Limited 1 Year)

US Machine Company

Our Local Commitment



Certified Safety (FDA Class 1)

Utilizing a CO2 laser integrated with a fully enclosed housing and a safety lid detection system, it fundamentally eliminates the risk of retinal damage—outperforming dangerous open-frame UV or Diode lasers. It ensures operator protection and gives you the most comprehensive protection.

Built to Last (Industrial-Grade Durability)

Engineered with scientifically rigorous electrical and mechanical designs, the machine undergoes long-term fatigue and abuse testing to ensure it is an industrial powerhouse and not a toy. Its robust industrial design ensures durability, maintaining reliability under high-volume, long-term use.



Local Support, Global Reach

Backed by a strong local U.S. technical support team, our service team provides quick assistance and rapid-response after-sales service. This ensures smooth operations and maximum uptime for your production line.



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